

No. of Functional Features Included in the Solution

Date	2 July 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Applicant Information Input	Allows users to enter applicant financial and personal details through a web form.	Flask UI	Done	High
2	Data Preprocessing Pipeline	Cleans data, handles missing values, performs feature encoding and scaling before prediction.	Data Preprocessing	Done	High
3	Credit Card Approval Prediction	Uses the trained machine learning model to predict approval or rejection.	ML Prediction Engine	Done	High
4	Multiple Model Training & Evaluation	Trains and compares Logistic Regression, Decision Tree, Random Forest, and XGBoost models to select the best performer.	Model Training	Done	High
5	Web-Based Prediction Dashboard	Displays prediction results instantly through an easy-to-use Flask web interface.	Flask Web Application	Done	High
6	IBM Watson Machine Learning Deployment	Deploys the trained model to IBM Cloud for scalable real-time predictions.	Cloud Deployment	Done	Medium
7	Input Validation & Error Handling	Validates user inputs and prevents invalid or incomplete data submission.	Backend Logic	Done	Medium
8	Model Saving & Loading	Saves the trained model using Pickle/Joblib and loads it for prediction without retraining.	Model Management	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Information Form, Prediction Result Dashboard
2	Backend / Logic	3	Data Preprocessing, ML Prediction Engine, Input Validation
3	Database / Storage	1	Model Storage using Pickle/Joblib
4	API / Integration	1	IBM Watson Machine Learning Cloud Deployment
5	Security / Authentication	1	Input Validation and Error Handling to prevent invalid data submission